Model Architecture: Qwen2.5-0.5B-Instruct

Generated by sg_modelload

October 24, 2026

Contents

1 Model Overview **2**

2 Mathematical Representation **2**
- 2.1 The model as a composition of layers . . . 2
- 2.2 The layers, top to bottom . . . 3
- 2.3 Neurons and attention heads . . . 3
- 2.4 Architecture and parameters . . . 4
- 2.5 Output representations . . . 4
- 2.6 Intermediate states and sub-network decomposition . . . 5
- 2.7 Local linear representation . . . 5
- 2.8 Summary of notation . . . 5

3 Forward Pass in Pseudocode **6**

4 Compact Architecture (repeated layers written once) **6**
- 4.1 Modules outside the repeated stacks . . . 6
- 4.2 Language-model decoder: `model.layers` . . . 6

5 What Each Layer Does **7**

6 Parameter Count **7**

A Full Layer-by-Layer Table **8**

B Per-Layer Parameter Count **13**

1 Model Overview

Model name	Qwen2.5-0.5B-Instruct
Transformers class	`Qwen2ForCausalLM`
Path	`/models/sayak_da_work/Qwen2.5-0.5B-Instruct`
Total parameters	494,032,768 (494.03M)
Weight dtype	bfloat16
Hidden size d	896
Decoder layers L	24
Query heads H	14
Key/value heads H_{kv}	2
MLP width d_{ff}	4,864
Vocabulary V	151,936
Max context length	32,768
RMSNorm ϵ	1e-06
Tied input/output embeddings	True
Activation	SiLU
RoPE base θ	1,000,000

How to read this document. Section 2 treats the model mathematically, top-down: first the whole network as one composition $\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E$ of layers, then a definition of every layer down to single neurons and attention heads, then parameters, outputs, intermediate states and the local linear response. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

Basic notation

B, T	batch size, sequence length (tokens)
d	hidden size (896)
V	vocabulary size (151,936)
$H,\ H_{kv},\ d_h$	query heads, key/value heads, size of one head
$W_*,\ b_*$	learned weight matrix / bias of the named projection
$\odot$, σ	element-wise product, logistic sigmoid
$f^{\circ n}$	f applied n times in a row ($f \circ f \circ \cdots \circ f$)
$g \circ f$	first f, then g (composition reads right to left)
RMSNorm(x)	$\gamma \odot x/\sqrt{\text{mean}(x^2)+\epsilon}$ (learned scale γ)
LN(x)	LayerNorm: $\gamma \odot (x-\mu)/\sigma + \beta$
RoPE	rotary position embedding (rotates query/key pairs by an angle set by position)

2 Mathematical Representation

We describe the model *top-down*. First the whole network is written as one composition of functions (Section 2.1). Then every function that appears in that composition is defined (Section 2.2), down to individual neurons and attention heads. The remaining subsections describe the parameters, the outputs, the intermediate states, and the local linear behaviour.

2.1 The model as a composition of layers

Definition 2.1 (Qwen2.5-0.5B-Instruct as an L-layer decoder)**.** The model is a function $f_\theta : \mathcal{X} \to \mathcal{Y}$ with parameters θ. It maps a sequence of T token ids $x \in \mathcal{X} = \{1, \ldots, V\}^T$ to one logit vector per position, $y \in \mathcal{Y} = \mathbb{R}^{T \times V}$ ($V = 151{,}936$). It is a composition of an embedding E, $L = 24$ decoder blocks $\mathcal{B}^{(\ell)}$ and an output head Ψ:

$$f_\theta \;=\; \Psi \circ \mathcal{B}^{(L)} \circ \mathcal{B}^{(L-1)} \circ \cdots \circ \mathcal{B}^{(2)} \circ \mathcal{B}^{(1)} \circ E \tag{1}$$

Every block maps $\mathbb{R}^{T \times d} \to \mathbb{R}^{T \times d}$ ($d = 896$), so the blocks can be stacked in any number. Block ℓ has its own parameters θ_ℓ and is stored in the checkpoint as `model.layers.`$(\ell - 1)$.

Definition 2.2 (Block types)**.** All L blocks have the same functional form $\mathcal{B}_{\text{F}}$ (Full (global) attention block). They differ only in their weights: $\mathcal{B}^{(\ell)} = \mathcal{B}_{\text{F}}(\,\cdot\,; \theta_\ell)$. Written out, with the type as a subscript:

$$f_\theta = \Psi \circ \mathcal{B}_{\text{F}}^{(24)} \circ \cdots \circ \mathcal{B}_{\text{F}}^{(3)} \circ \mathcal{B}_{\text{F}}^{(2)} \circ \mathcal{B}_{\text{F}}^{(1)} \circ E \tag{2}$$

Grouping the repetitions with composition powers ($\mathcal{B}^{\circ n} = n$ consecutive blocks of the same type, each with its own weights) gives the compact form

$$f_\theta = \Psi \circ \left(\mathcal{B}_{\mathrm{F}}^{\circ 24}\right) \circ E \tag{3}$$

Layer pattern.

Symbol	Block type	Count	Params / block	Checkpoint indices
$\mathcal{B}_{\mathrm{F}}$	Full (global) attention block	24	14,912,384	0–23

Language-model decoder (24 layers). layers 0–23: $[\mathcal{B}_{\mathrm{F}}]$ repeated 24× (checkpoint indices, i.e. $\ell - 1$).

2.2 The layers, top to bottom

Every function in the composition above is defined here. Only the attention (or linear-attention) sub-layer mixes information *between* positions t. Every other sub-layer (norms, feed-forward networks, projections) acts on each token vector $h_t \in \mathbb{R}^d$ independently, e.g. $\mathrm{FFN}(h)_t = \mathrm{FFN}(h_t)$. All weights follow the convention $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, so a projection of a token vector is Wu (as in an affine layer $f_i(u) = A^{(i)}u + b^{(i)}$).

Definition 2.3 (Token embedding)**.** $E : \{1, \ldots, V\}^T \to \mathbb{R}^{T \times d}$ looks up one row of the embedding matrix $W_E \in \mathbb{R}^{V \times d}$ per token:

$$E(x)_t = W_E[x_t, :], \qquad t = 1, \ldots, T. \tag{4}$$

Definition 2.4 (Normalisation)**.** $\mathrm{RMSNorm}(u) = \gamma \odot u / \sqrt{\frac{1}{d}\sum_i u_i^2 + \epsilon}$, with a learned scale $\gamma \in \mathbb{R}^d$, applied to each token vector separately. It fixes the scale of a vector without changing its direction. Every norm in the model has its own γ, distinguished by a subscript (e.g. $\mathrm{RMSNorm}_1$, $\mathrm{RMSNorm}_{\text{final}}$).

Definition 2.5 (Block $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block)**.** A decoder block of this type maps $h \in \mathbb{R}^{T \times 896}$ to the same space. It is a composition of 2 residual sub-layers, each of the form $(\mathrm{id} + g)(h) = h + g(h)$:

$$\mathcal{B}_{\mathrm{F}} = (\mathrm{id} + \mathrm{FFN} \circ \mathrm{RMSNorm}_2) \circ (\mathrm{id} + \mathrm{Attn} \circ \mathrm{RMSNorm}_1) \tag{5}$$

Step by step, the sub-layers are:

$$\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{RMSNorm}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a)) && \text{pre-norm + residual}\\
Q &= W_Q\, u + b_Q && H = 14 \text{ heads of size } d_h = 64\\
K,\ V &= W_K\, u + b_K, \quad W_V\, u + b_V && H_{kv} = 2 \text{ key/value heads}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\, Q_h K_{g(h)}^\top + M && \text{mask } M\text{: causal}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\, V_{g(h)} &&\\
\mathrm{Attn}(u) &= W_O\,[O_1; \ldots; O_H] &&\\
\mathrm{FFN}(u) &= W_{\text{down}}\big(\mathrm{SiLU}(W_{\text{gate}}\, u) \odot W_{\text{up}}\, u\big) && \text{gated MLP, width 4864}
\end{aligned}$$

Parameters of one block ($|\theta_\ell| = 14{,}912{,}384$): $W_Q \in \mathbb{R}^{896\times896}$, $W_K \in \mathbb{R}^{128\times896}$, $W_V \in \mathbb{R}^{128\times896}$, $W_O \in \mathbb{R}^{896\times896}$, $W_{\text{gate}} \in \mathbb{R}^{4864\times896}$, $W_{\text{up}} \in \mathbb{R}^{4864\times896}$, $W_{\text{down}} \in \mathbb{R}^{896\times4864}$, plus 2,944 parameters in norm scales, biases and other vectors. This type is used 24 times.

Definition 2.6 (Output head)**.** $\Psi : \mathbb{R}^{T\times d} \to \mathbb{R}^{T\times V}$ normalises the last hidden state and scores every vocabulary token, re-using the embedding matrix (tied weights):

$$\Psi(h)_t = W_E^\top\ \mathrm{RMSNorm}_{\text{final}}(h)_t \tag{6}$$

2.3 Neurons and attention heads

Definition 2.7 (Neuron, pre-activation value and state)**.** The feed-forward sub-layer of block ℓ contains $d_{ff}^{(\ell)} = 4864$ neurons. Neuron j of layer ℓ is the scalar function of a normalised token vector $u \in \mathbb{R}^d$

$$\eta_j^{(\ell)}(u) = \mathrm{SiLU}\left(W_{\text{gate},j}^{(\ell)} \cdot u\right)\left(W_{\text{up},j}^{(\ell)} \cdot u\right), \tag{7}$$

where $W_{\text{gate},j}$ is the j-th row of W_{gate}. The feed-forward output is the sum of the neuron outputs, each written along its own column of W_{down}:

$$\text{FFN}_\ell(u) = \sum_{j=1}^{d_{ff}^{(\ell)}} \eta_j^{(\ell)}(u)\; W_{\text{down}}^{(\ell)}[:, j]. \tag{8}$$

Across its L blocks the network has $N = \sum_{\ell=1}^{L} d_{ff}^{(\ell)} = 116{,}736$ feed-forward neurons. For input x at position t, with $u_t^{(\ell)}(x)$ the normalised input of the feed-forward sub-layer:

- $\mathcal{V}(\eta; x, t) = W_{\text{gate},j}^{(\ell)} \cdot u_t^{(\ell)}(x) \in \mathbb{R}$ is the **pre-activation value** of neuron η (the argument of the non-linearity).
- $\mathcal{S}(\eta; x, t)$ is its **activation state**: *active* if $\mathcal{V} > 0$ and *inactive* if $\mathcal{V} \leq 0$. The non-linearity is smooth, unlike ReLU, so an inactive neuron's output is strongly suppressed rather than exactly zero.

Definition 2.8 (Attention head and attention pattern)**.** Head h of an attention layer ℓ reads from the whole sequence through its attention pattern $P_h^{(\ell)}(x) \in \mathbb{R}^{T \times T}$ (row t = how much position t attends to each earlier position) and writes through its own $d \times d_h$ slice $W_{O,h}^{(\ell)}$ of W_O:

$$\text{Attn}_\ell(u) = \sum_{h=1}^{H} P_h^{(\ell)}\, V_{g(h)}\, W_{O,h}^{(\ell)\top}. \tag{9}$$

$g(h)$ is the key/value head shared by query head h (grouped-query attention). The pattern P_h plays the role that the activation state plays for neurons: once P_h is fixed, the head is linear in its values. The network has 24 attention layers $\times$ $H = 14$ heads $= 336$ attention heads.

2.4 Architecture and parameters

Definition 2.9 (Architecture and parameters)**.** The **architecture** is the tuple of sizes and the layer-type sequence:

$$\mathcal{A} = (V, d, L, c, H, H_{kv}, d_h, d_{ff}) = (151{,}936,\ 896,\ 24,\ (\text{F}^{24}),\ 14,\ 2,\ 64,\ 4864). \tag{10}$$

The **parameters** are $\theta = \{W_E,\ \theta_1, \ldots, \theta_L,\ \gamma_{\text{final}}\}$ (W_E is shared with the output head). Grouping the blocks by type,

$$\begin{aligned} |\theta| &= \sum_{\text{types } c} n_c\, |\theta_c| + |\theta_{\text{rest}}| \\ &= 24 \times 14{,}912{,}384 + 136{,}135{,}552 \\ &= 494{,}032{,}768, \end{aligned}$$

where n_c is how often type c occurs and θ_{rest} holds everything outside the repeated stacks (embedding, final norm, …; Table 3).

2.5 Output representations

Definition 2.10 (Output representations)**.** For an input sequence x, the output can be read at three levels:

1. **Logits:** $y = f_\theta(x) \in \mathbb{R}^{T \times V}$, one score per vocabulary token at every position.
2. **Next-token distribution:** $p(x)_{t,k} = \dfrac{\exp(y_{t,k})}{\sum_{j=1}^{V} \exp(y_{t,j})} = \Pr\big(x_{t+1} = k \mid x_1, \ldots, x_t\big)$.
3. **Predicted next token:** $z(x) = \text{argmax}_{k \in \{1,\ldots,V\}}\, y_{T,k}$ (greedy decoding). Generation appends $z(x)$ (or a sample from $p(x)_T$) to x and applies f_θ again; in practice the keys and values of earlier positions are cached.

The model is **causal**: y_t depends only on $x_1, \ldots, x_t$, because every attention mask hides later positions. Training minimises the cross-entropy $-\sum_t \log p(x)_{t, x_{t+1}}$.

2.6 Intermediate states and sub-network decomposition

To study what happens at one layer $\ell \in \{1, \dots, L\}$, split the composition into a prefix, the layer itself and a suffix:

$$f_\theta = \underbrace{\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(\ell+1)}}_{G_{\ell+1}\ \text{(suffix)}} \circ \underbrace{\mathcal{B}^{(\ell)}}_{\text{layer } \ell} \circ \underbrace{\mathcal{B}^{(\ell-1)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E}_{F_{\ell-1}\ \text{(prefix)}} \tag{11}$$

- **Prefix** $F_{\ell-1} : \mathcal{X} \to \mathbb{R}^{T \times d}$ maps the tokens to the hidden state entering layer ℓ ($F_0 = E$).
- **Hidden state** (residual stream) after layer ℓ: $h^{(\ell)} = F_\ell(x) = \mathcal{B}^{(\ell)}(h^{(\ell-1)}) \in \mathbb{R}^{T \times d}$, with $h^{(0)} = E(x)$.
- **Inside layer ℓ:** the attention write $a^{(\ell)} = \mathrm{Attn}_\ell\big(\mathrm{RMSNorm}_1(h^{(\ell-1)})\big)$, the mid-block state $\hat{h}^{(\ell)} = h^{(\ell-1)} + a^{(\ell)}$, the feed-forward input $u^{(\ell)} = \mathrm{RMSNorm}_2(\hat{h}^{(\ell)})$, the pre-activations $\hat{x}^{(\ell)} = u^{(\ell)} W_{\text{gate}}^{(\ell)\top} \in \mathbb{R}^{T \times d_{ff}}$ and the neuron outputs $\eta^{(\ell)}(u^{(\ell)}) \in \mathbb{R}^{T \times d_{ff}}$, and the feed-forward write $m^{(\ell)} = \mathrm{FFN}_\ell(u^{(\ell)})$.
- **Suffix** $G_{\ell+1} : \mathbb{R}^{T \times d} \to \mathbb{R}^{T \times V}$ maps $h^{(\ell)}$ to the logits ($G_{L+1} = \Psi$).

Residual stream. Because every sub-layer only *adds* to the hidden state, unrolling the composition gives the final hidden state as the embedding plus everything the layers wrote:

$$h^{(L)} = h^{(0)} + \sum_{\ell=1}^{L} \big(a^{(\ell)} + m^{(\ell)}\big) \tag{12}$$

So $f_\theta(x) = \Psi(h^{(L)})$: each layer reads the current stream and adds a correction to it.

2.7 Local linear representation

A ReLU network is piecewise affine: inside a polytope cell it collapses to one matrix. This model is not, because softmax, SiLU and the normalisations are smooth. Its local behaviour is described by the first-order (Jacobian) linearisation instead. Tokens are discrete, so we perturb the embedded input $h^{(0)}$.

Definition 2.11 (Local linear response)**.** Let $J_\ell = D\mathcal{B}^{(\ell)}(h^{(\ell-1)})$ be the Jacobian of block ℓ at the current hidden state (tensors flattened to vectors). The chain rule applied to the residual sub-layers $(\mathrm{id} + g_{\ell,i})$ gives

$$J_\ell = (I + Dg_{\ell,2})(I + Dg_{\ell,1}), \tag{13}$$

where $g_{\ell,1}$ is the attention branch, $g_{\ell,2}$ the feed-forward branch and each Dg is taken at that branch's input. For a small perturbation δ of $h^{(0)}$, the response of the network decomposed around layer ℓ is

$$f_\theta(h^{(0)} + \delta) - f_\theta(h^{(0)}) \approx G_x^{(\ell+1)} \cdot J_\ell \cdot F_x^{(\ell-1)} \cdot \delta, \tag{14}$$

with $F_x^{(\ell-1)} = J_{\ell-1} \cdots J_1$ (prefix) and $G_x^{(\ell+1)} = J_\Psi\, J_L \cdots J_{\ell+1}$ (suffix). Multiplying out the products of $(I + Dg)$ terms writes $J_\Psi J_L \cdots J_1$ as a sum over *paths*: the identity path straight from embedding to head, plus one term for every subset of sub-layers a signal passes through.

2.8 Summary of notation

Symbol	Meaning
$f_\theta : \mathcal{X} \to \mathcal{Y}$	the whole model, $\{1..V\}^T \to \mathbb{R}^{T \times V}$
$E,\ \Psi$	embedding and output head
$\mathcal{B}^{(\ell)},\ \theta_\ell$	block ℓ and its parameters, $\ell = 1, \dots, L$
$\mathcal{B}_c,\ c(\ell)$	block type c; type of layer ℓ
$\mathcal{B}^{\circ n}$	n consecutive blocks of one type (each with its own weights)
$(\mathrm{id} + g)$	residual sub-layer $h \mapsto h + g(h)$
$h^{(\ell)}$	hidden state (residual stream) after layer ℓ; $h^{(0)}$ = embeddings
$a^{(\ell)},\ m^{(\ell)}$	what the attention / feed-forward sub-layer of layer ℓ adds to the stream
$\eta_j^{(\ell)}$	j-th feed-forward neuron of layer ℓ
$\mathcal{V}(\eta; x, t),\ \mathcal{S}(\eta; x, t)$	pre-activation value and active/inactive state of a neuron
$P_h^{(\ell)}$	attention pattern of head h in layer ℓ ($T \times T$)

continued on next page

Symbol	Meaning
$F_{\ell-1},\ G_{\ell+1}$	prefix (embedding to layer $\ell-1$) and suffix (layer $\ell+1$ to logits)
$J_\ell,\ F_x^{(\ell-1)},\ G_x^{(\ell+1)}$	Jacobian of block ℓ, and of the prefix / suffix at input x
$p(x),\ z(x)$	next-token distribution and greedy next-token prediction

3 Forward Pass in Pseudocode

Algorithm 1 Qwen2.5-0.5B-Instruct: full forward pass

Require: token ids x of shape $B \times T$

$h \leftarrow E(x)$ ▷ $h^{(0)}$: one vector per token, $B{\times}T{\times}896$

for $\ell \leftarrow 1$ **to** 24 **do** ▷ 24 blocks of type $\mathcal{B}_{\mathrm{F}}$

  $h \leftarrow \mathcal{B}_{\mathrm{F}}(h;\, \theta_\ell)$ ▷ Full (global) attention block; result is $h^{(\ell)}$

end for

$h \leftarrow \mathrm{RMSNorm}_{\mathrm{final}}(h)$ ▷ $h = h^{(L)}$; this and the next step are Ψ

$y \leftarrow h\, W_E^\top$ ▷ logits $B{\times}T{\times}151,936$, embedding matrix re-used

return y ▷ softmax(y) = next-token probabilities

Algorithm 2 Block $\mathcal{B}_{\mathrm{F}}(h)$ — Full (global) attention block

Require: h of shape $B \times T \times 896$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection

$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector

token mixing: multi-head self-attention

$Q \leftarrow uW_Q^\top + b_Q$ ▷ $B{\times}T{\times}14{\times}64$ (14 query heads)

$K \leftarrow uW_K^\top + b_K;\ V \leftarrow uW_V^\top + b_V$ ▷ $B{\times}T{\times}2{\times}64$ (2 key/value heads, each shared by 7 query heads)

$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs

$P \leftarrow \mathrm{softmax}(s\, QK^\top + M)$ ▷ mask: causal: a token sees only earlier tokens

$O \leftarrow PV$ ▷ $B{\times}T{\times}14{\times}64$

$m \leftarrow O\, W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}896$

$h \leftarrow r + m$ ▷ add residual

channel mixing: feed-forward network (each token independently)

$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$

$m \leftarrow \big(\mathrm{SiLU}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big) W_{\mathrm{down}}^\top$ ▷ expand to 4864, gate, shrink back to 896

$h \leftarrow r + m$ ▷ add residual

return h

4 Compact Architecture (repeated layers written once)

4.1 Modules outside the repeated stacks

Table 3: Weights that appear once

Weight	Type	Shape	Params
`model.embed_tokens.weight`	Embedding	151936 × 896	136,134,656
`model.norm.weight`	Qwen2RMSNorm	896	896
`lm_head.weight` (tied to `model.embed_tokens.weight`)	Linear	151936 × 896	0 (shared)

4.2 Language-model decoder: `model.layers`

Table 4: $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block — layers 0–23; weight names are relative to `model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	896×896	802,816
`self_attn.q_proj.bias`	Linear	896	896
`self_attn.k_proj.weight`	Linear	128×896	114,688
`self_attn.k_proj.bias`	Linear	128	128
`self_attn.v_proj.weight`	Linear	128×896	114,688
`self_attn.v_proj.bias`	Linear	128	128
`self_attn.o_proj.weight`	Linear	896×896	802,816
`mlp.gate_proj.weight`	Linear	4864×896	4,358,144
`mlp.up_proj.weight`	Linear	4864×896	4,358,144
`mlp.down_proj.weight`	Linear	896×4864	4,358,144
`input_layernorm.weight`	Qwen2RMSNorm	896	896
`post_attention_layernorm.weight`	Qwen2RMSNorm	896	896
Per block			**14,912,384**
$\times$ 24 blocks			**357,897,216**

5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

Layer (appears as)	What it does
LANGUAGE MODEL	
Token embedding `embed_tokens`	Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$).
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?".
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
Gate projection (gate_proj) `mlp.gate_proj`	Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit.
Up projection (up_proj) `mlp.up_proj`	Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU).
Down projection (down_proj) `mlp.down_proj`	Shrinks the gated hidden vector back to the model width so it can be added to the residual stream.
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	Despite the name, this is the pre-MLP norm: RMSNorm on the residual stream after attention, feeding the MLP.
Final norm `norm`	RMSNorm applied once after the last decoder layer, just before the LM head.

6 Parameter Count

Table 6: Parameters by component

Component	Parameters	% of total
MLP / MoE	313,786,368	63.5%
Token embedding	136,134,656	27.6%
Attention / token mixer	44,067,840	8.9%
Norms	43,904	0.0%
Total	**494,032,768**	**100.0%**

Table 7: Parameters by block type

Block type	Count	Params / block	Total
$\mathcal{B}_{\mathrm{F}}$ Full (global) attention block	24	14,912,384	357,897,216
Modules outside stacks	1	–	136,135,552
Total			**494,032,768**

Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

Layer path	Type	Shape	Dtype	Params
`lm_head.weight` *(tied)*	Linear	151936 × 896	bfloat16	136,134,656
`model`	Qwen2Model	–	–	–
`model.embed_tokens.weight`	Embedding	151936 × 896	bfloat16	136,134,656
`model.layers.0`	Qwen2DecoderLayer	–	–	–
`model.layers.0.input_layernorm.weight`	Qwen2RMSNorm	896	bfloat16	896
`model.layers.0.mlp`	Qwen2MLP	–	–	–
`model.layers.0.mlp.down_proj.weight`	Linear	896 × 4864	bfloat16	4,358,144
`model.layers.0.mlp.gate_proj.weight`	Linear	4864 × 896	bfloat16	4,358,144
`model.layers.0.mlp.up_proj.weight`	Linear	4864 × 896	bfloat16	4,358,144
`model.layers.0.post_attention_layernorm.weight`	Qwen2RMSNorm	896	bfloat16	896
`model.layers.0.self_attn`	Qwen2Attention	–	–	–
`model.layers.0.self_attn.k_proj.bias`	Linear	128	bfloat16	128
`model.layers.0.self_attn.k_proj.weight`	Linear	128 × 896	bfloat16	114,688
`model.layers.0.self_attn.o_proj.weight`	Linear	896 × 896	bfloat16	802,816
`model.layers.0.self_attn.q_proj.bias`	Linear	896	bfloat16	896
`model.layers.0.self_attn.q_proj.weight`	Linear	896 × 896	bfloat16	802,816
`model.layers.0.self_attn.v_proj.bias`	Linear	128	bfloat16	128
`model.layers.0.self_attn.v_proj.weight`	Linear	128 × 896	bfloat16	114,688
`model.layers.1`	Qwen2DecoderLayer	–	–	–
`model.layers.1.input_layernorm.weight`	Qwen2RMSNorm	896	bfloat16	896
`model.layers.1.mlp`	Qwen2MLP	–	–	–
`model.layers.1.mlp.down_proj.weight`	Linear	896 × 4864	bfloat16	4,358,144
`model.layers.1.mlp.gate_proj.weight`	Linear	4864 × 896	bfloat16	4,358,144
`model.layers.1.mlp.up_proj.weight`	Linear	4864 × 896	bfloat16	4,358,144
`model.layers.1.post_attention_layernorm.weight`	Qwen2RMSNorm	896	bfloat16	896
`model.layers.1.self_attn`	Qwen2Attention	–	–	–
`model.layers.1.self_attn.k_proj.bias`	Linear	128	bfloat16	128
`model.layers.1.self_attn.k_proj.weight`	Linear	128 × 896	bfloat16	114,688
`model.layers.1.self_attn.o_proj.weight`	Linear	896 × 896	bfloat16	802,816
`model.layers.1.self_attn.q_proj.bias`	Linear	896	bfloat16	896
`model.layers.1.self_attn.q_proj.weight`	Linear	896 × 896	bfloat16	802,816
`model.layers.1.self_attn.v_proj.bias`	Linear	128	bfloat16	128
`model.layers.1.self_attn.v_proj.weight`	Linear	128 × 896	bfloat16	114,688
`model.layers.2`	Qwen2DecoderLayer	–	–	–
`model.layers.2.input_layernorm.weight`	Qwen2RMSNorm	896	bfloat16	896
`model.layers.2.mlp`	Qwen2MLP	–	–	–
`model.layers.2.mlp.down_proj.weight`	Linear	896 × 4864	bfloat16	4,358,144
`model.layers.2.mlp.gate_proj.weight`	Linear	4864 × 896	bfloat16	4,358,144
`model.layers.2.mlp.up_proj.weight`	Linear	4864 × 896	bfloat16	4,358,144
`model.layers.2.post_attention_layernorm.weight`	Qwen2RMSNorm	896	bfloat16	896
`model.layers.2.self_attn`	Qwen2Attention	–	–	–
`model.layers.2.self_attn.k_proj.bias`	Linear	128	bfloat16	128
`model.layers.2.self_attn.k_proj.weight`	Linear	128 × 896	bfloat16	114,688

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Layer path	Type	Shape	Dtype	Params
model.layers.2.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.2.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.2.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.2.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.2.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.3	Qwen2DecoderLayer	–	–	–
model.layers.3.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.3.mlp	Qwen2MLP	–	–	–
model.layers.3.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.3.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.3.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.3.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.3.self_attn	Qwen2Attention	–	–	–
model.layers.3.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.3.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.3.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.3.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.3.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.3.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.3.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.4	Qwen2DecoderLayer	–	–	–
model.layers.4.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.4.mlp	Qwen2MLP	–	–	–
model.layers.4.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.4.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.4.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.4.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.4.self_attn	Qwen2Attention	–	–	–
model.layers.4.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.4.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.4.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.4.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.4.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.4.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.4.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.5	Qwen2DecoderLayer	–	–	–
model.layers.5.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.5.mlp	Qwen2MLP	–	–	–
model.layers.5.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.5.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.5.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.5.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.5.self_attn	Qwen2Attention	–	–	–
model.layers.5.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.5.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.5.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.5.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.5.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.5.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.5.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.6	Qwen2DecoderLayer	–	–	–
model.layers.6.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.6.mlp	Qwen2MLP	–	–	–
model.layers.6.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.6.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.6.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.6.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.6.self_attn	Qwen2Attention	–	–	–
model.layers.6.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.6.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.6.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.6.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.6.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.6.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.6.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.7	Qwen2DecoderLayer	–	–	–
model.layers.7.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.7.mlp	Qwen2MLP	–	–	–
model.layers.7.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.7.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144

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Layer path	Type	Shape	Dtype	Params
model.layers.7.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.7.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.7.self_attn	Qwen2Attention	–	–	–
model.layers.7.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.7.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.7.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.7.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.7.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.7.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.7.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.8	Qwen2DecoderLayer	–	–	–
model.layers.8.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.8.mlp	Qwen2MLP	–	–	–
model.layers.8.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.8.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.8.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.8.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.8.self_attn	Qwen2Attention	–	–	–
model.layers.8.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.8.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.8.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.8.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.8.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.8.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.8.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.9	Qwen2DecoderLayer	–	–	–
model.layers.9.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.9.mlp	Qwen2MLP	–	–	–
model.layers.9.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.9.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.9.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.9.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.9.self_attn	Qwen2Attention	–	–	–
model.layers.9.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.9.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.9.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.9.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.9.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.9.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.9.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.10	Qwen2DecoderLayer	–	–	–
model.layers.10.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.10.mlp	Qwen2MLP	–	–	–
model.layers.10.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.10.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.10.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.10.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.10.self_attn	Qwen2Attention	–	–	–
model.layers.10.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.10.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.10.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.10.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.10.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.10.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.10.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.11	Qwen2DecoderLayer	–	–	–
model.layers.11.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.11.mlp	Qwen2MLP	–	–	–
model.layers.11.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.11.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.11.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.11.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.11.self_attn	Qwen2Attention	–	–	–
model.layers.11.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.11.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.11.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.11.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.11.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.11.self_attn.v_proj.bias	Linear	128	bfloat16	128

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Layer path	Type	Shape	Dtype	Params
model.layers.11.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.12	Qwen2DecoderLayer	–	–	–
model.layers.12.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.12.mlp	Qwen2MLP	–	–	–
model.layers.12.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.12.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.12.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.12.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.12.self_attn	Qwen2Attention	–	–	–
model.layers.12.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.12.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.12.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.12.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.12.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.12.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.12.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.13	Qwen2DecoderLayer	–	–	–
model.layers.13.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.13.mlp	Qwen2MLP	–	–	–
model.layers.13.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.13.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.13.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.13.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.13.self_attn	Qwen2Attention	–	–	–
model.layers.13.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.13.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.13.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.13.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.13.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.13.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.13.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.14	Qwen2DecoderLayer	–	–	–
model.layers.14.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.14.mlp	Qwen2MLP	–	–	–
model.layers.14.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.14.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.14.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.14.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.14.self_attn	Qwen2Attention	–	–	–
model.layers.14.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.14.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.14.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.14.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.14.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.14.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.14.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.15	Qwen2DecoderLayer	–	–	–
model.layers.15.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.15.mlp	Qwen2MLP	–	–	–
model.layers.15.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.15.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.15.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.15.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.15.self_attn	Qwen2Attention	–	–	–
model.layers.15.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.15.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.15.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.15.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.15.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.15.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.15.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.16	Qwen2DecoderLayer	–	–	–
model.layers.16.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.16.mlp	Qwen2MLP	–	–	–
model.layers.16.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.16.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.16.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.16.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.16.self_attn	Qwen2Attention	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.layers.16.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.16.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.16.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.16.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.16.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.16.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.16.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.17	Qwen2DecoderLayer	–	–	–
model.layers.17.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.17.mlp	Qwen2MLP	–	–	–
model.layers.17.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.17.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.17.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.17.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.17.self_attn	Qwen2Attention	–	–	–
model.layers.17.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.17.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.17.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.17.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.17.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.17.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.17.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.18	Qwen2DecoderLayer	–	–	–
model.layers.18.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.18.mlp	Qwen2MLP	–	–	–
model.layers.18.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.18.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.18.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.18.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.18.self_attn	Qwen2Attention	–	–	–
model.layers.18.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.18.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.18.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.18.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.18.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.18.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.18.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.19	Qwen2DecoderLayer	–	–	–
model.layers.19.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.19.mlp	Qwen2MLP	–	–	–
model.layers.19.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.19.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.19.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.19.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.19.self_attn	Qwen2Attention	–	–	–
model.layers.19.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.19.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.19.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.19.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.19.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.19.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.19.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.20	Qwen2DecoderLayer	–	–	–
model.layers.20.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.20.mlp	Qwen2MLP	–	–	–
model.layers.20.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.20.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.20.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.20.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.20.self_attn	Qwen2Attention	–	–	–
model.layers.20.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.20.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.20.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.20.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.20.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.20.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.20.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.21	Qwen2DecoderLayer	–	–	–
model.layers.21.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.21.mlp	Qwen2MLP	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.layers.21.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.21.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.21.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.21.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.21.self_attn	Qwen2Attention	–	–	–
model.layers.21.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.21.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.21.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.21.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.21.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.21.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.21.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.22	Qwen2DecoderLayer	–	–	–
model.layers.22.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.22.mlp	Qwen2MLP	–	–	–
model.layers.22.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.22.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.22.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.22.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.22.self_attn	Qwen2Attention	–	–	–
model.layers.22.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.22.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.22.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.22.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.22.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.22.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.22.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.23	Qwen2DecoderLayer	–	–	–
model.layers.23.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.23.mlp	Qwen2MLP	–	–	–
model.layers.23.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.23.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.23.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.23.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.23.self_attn	Qwen2Attention	–	–	–
model.layers.23.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.23.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.23.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.23.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.23.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.23.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.23.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.norm.weight	Qwen2RMSNorm	896	bfloat16	896

Appendix B Per-Layer Parameter Count

Module	Params	Running total
model.embed_tokens	136,134,656	136,134,656
model.layers.0.input_layernorm	896	136,135,552
model.layers.0.mlp.down_proj	4,358,144	140,493,696
model.layers.0.mlp.gate_proj	4,358,144	144,851,840
model.layers.0.mlp.up_proj	4,358,144	149,209,984
model.layers.0.post_attention_layernorm	896	149,210,880
model.layers.0.self_attn.k_proj	114,816	149,325,696
model.layers.0.self_attn.o_proj	802,816	150,128,512
model.layers.0.self_attn.q_proj	803,712	150,932,224
model.layers.0.self_attn.v_proj	114,816	151,047,040
model.layers.1.input_layernorm	896	151,047,936
model.layers.1.mlp.down_proj	4,358,144	155,406,080
model.layers.1.mlp.gate_proj	4,358,144	159,764,224
model.layers.1.mlp.up_proj	4,358,144	164,122,368
model.layers.1.post_attention_layernorm	896	164,123,264
model.layers.1.self_attn.k_proj	114,816	164,238,080

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Module	Params	Running total
model.layers.1.self_attn.o_proj	802,816	165,040,896
model.layers.1.self_attn.q_proj	803,712	165,844,608
model.layers.1.self_attn.v_proj	114,816	165,959,424
model.layers.2.input_layernorm	896	165,960,320
model.layers.2.mlp.down_proj	4,358,144	170,318,464
model.layers.2.mlp.gate_proj	4,358,144	174,676,608
model.layers.2.mlp.up_proj	4,358,144	179,034,752
model.layers.2.post_attention_layernorm	896	179,035,648
model.layers.2.self_attn.k_proj	114,816	179,150,464
model.layers.2.self_attn.o_proj	802,816	179,953,280
model.layers.2.self_attn.q_proj	803,712	180,756,992
model.layers.2.self_attn.v_proj	114,816	180,871,808
model.layers.3.input_layernorm	896	180,872,704
model.layers.3.mlp.down_proj	4,358,144	185,230,848
model.layers.3.mlp.gate_proj	4,358,144	189,588,992
model.layers.3.mlp.up_proj	4,358,144	193,947,136
model.layers.3.post_attention_layernorm	896	193,948,032
model.layers.3.self_attn.k_proj	114,816	194,062,848
model.layers.3.self_attn.o_proj	802,816	194,865,664
model.layers.3.self_attn.q_proj	803,712	195,669,376
model.layers.3.self_attn.v_proj	114,816	195,784,192
model.layers.4.input_layernorm	896	195,785,088
model.layers.4.mlp.down_proj	4,358,144	200,143,232
model.layers.4.mlp.gate_proj	4,358,144	204,501,376
model.layers.4.mlp.up_proj	4,358,144	208,859,520
model.layers.4.post_attention_layernorm	896	208,860,416
model.layers.4.self_attn.k_proj	114,816	208,975,232
model.layers.4.self_attn.o_proj	802,816	209,778,048
model.layers.4.self_attn.q_proj	803,712	210,581,760
model.layers.4.self_attn.v_proj	114,816	210,696,576
model.layers.5.input_layernorm	896	210,697,472
model.layers.5.mlp.down_proj	4,358,144	215,055,616
model.layers.5.mlp.gate_proj	4,358,144	219,413,760
model.layers.5.mlp.up_proj	4,358,144	223,771,904
model.layers.5.post_attention_layernorm	896	223,772,800
model.layers.5.self_attn.k_proj	114,816	223,887,616
model.layers.5.self_attn.o_proj	802,816	224,690,432
model.layers.5.self_attn.q_proj	803,712	225,494,144
model.layers.5.self_attn.v_proj	114,816	225,608,960
model.layers.6.input_layernorm	896	225,609,856
model.layers.6.mlp.down_proj	4,358,144	229,968,000
model.layers.6.mlp.gate_proj	4,358,144	234,326,144
model.layers.6.mlp.up_proj	4,358,144	238,684,288
model.layers.6.post_attention_layernorm	896	238,685,184
model.layers.6.self_attn.k_proj	114,816	238,800,000
model.layers.6.self_attn.o_proj	802,816	239,602,816
model.layers.6.self_attn.q_proj	803,712	240,406,528
model.layers.6.self_attn.v_proj	114,816	240,521,344
model.layers.7.input_layernorm	896	240,522,240
model.layers.7.mlp.down_proj	4,358,144	244,880,384
model.layers.7.mlp.gate_proj	4,358,144	249,238,528
model.layers.7.mlp.up_proj	4,358,144	253,596,672
model.layers.7.post_attention_layernorm	896	253,597,568
model.layers.7.self_attn.k_proj	114,816	253,712,384
model.layers.7.self_attn.o_proj	802,816	254,515,200
model.layers.7.self_attn.q_proj	803,712	255,318,912
model.layers.7.self_attn.v_proj	114,816	255,433,728
model.layers.8.input_layernorm	896	255,434,624

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Module	Params	Running total
model.layers.8.mlp.down_proj	4,358,144	259,792,768
model.layers.8.mlp.gate_proj	4,358,144	264,150,912
model.layers.8.mlp.up_proj	4,358,144	268,509,056
model.layers.8.post_attention_layernorm	896	268,509,952
model.layers.8.self_attn.k_proj	114,816	268,624,768
model.layers.8.self_attn.o_proj	802,816	269,427,584
model.layers.8.self_attn.q_proj	803,712	270,231,296
model.layers.8.self_attn.v_proj	114,816	270,346,112
model.layers.9.input_layernorm	896	270,347,008
model.layers.9.mlp.down_proj	4,358,144	274,705,152
model.layers.9.mlp.gate_proj	4,358,144	279,063,296
model.layers.9.mlp.up_proj	4,358,144	283,421,440
model.layers.9.post_attention_layernorm	896	283,422,336
model.layers.9.self_attn.k_proj	114,816	283,537,152
model.layers.9.self_attn.o_proj	802,816	284,339,968
model.layers.9.self_attn.q_proj	803,712	285,143,680
model.layers.9.self_attn.v_proj	114,816	285,258,496
model.layers.10.input_layernorm	896	285,259,392
model.layers.10.mlp.down_proj	4,358,144	289,617,536
model.layers.10.mlp.gate_proj	4,358,144	293,975,680
model.layers.10.mlp.up_proj	4,358,144	298,333,824
model.layers.10.post_attention_layernorm	896	298,334,720
model.layers.10.self_attn.k_proj	114,816	298,449,536
model.layers.10.self_attn.o_proj	802,816	299,252,352
model.layers.10.self_attn.q_proj	803,712	300,056,064
model.layers.10.self_attn.v_proj	114,816	300,170,880
model.layers.11.input_layernorm	896	300,171,776
model.layers.11.mlp.down_proj	4,358,144	304,529,920
model.layers.11.mlp.gate_proj	4,358,144	308,888,064
model.layers.11.mlp.up_proj	4,358,144	313,246,208
model.layers.11.post_attention_layernorm	896	313,247,104
model.layers.11.self_attn.k_proj	114,816	313,361,920
model.layers.11.self_attn.o_proj	802,816	314,164,736
model.layers.11.self_attn.q_proj	803,712	314,968,448
model.layers.11.self_attn.v_proj	114,816	315,083,264
model.layers.12.input_layernorm	896	315,084,160
model.layers.12.mlp.down_proj	4,358,144	319,442,304
model.layers.12.mlp.gate_proj	4,358,144	323,800,448
model.layers.12.mlp.up_proj	4,358,144	328,158,592
model.layers.12.post_attention_layernorm	896	328,159,488
model.layers.12.self_attn.k_proj	114,816	328,274,304
model.layers.12.self_attn.o_proj	802,816	329,077,120
model.layers.12.self_attn.q_proj	803,712	329,880,832
model.layers.12.self_attn.v_proj	114,816	329,995,648
model.layers.13.input_layernorm	896	329,996,544
model.layers.13.mlp.down_proj	4,358,144	334,354,688
model.layers.13.mlp.gate_proj	4,358,144	338,712,832
model.layers.13.mlp.up_proj	4,358,144	343,070,976
model.layers.13.post_attention_layernorm	896	343,071,872
model.layers.13.self_attn.k_proj	114,816	343,186,688
model.layers.13.self_attn.o_proj	802,816	343,989,504
model.layers.13.self_attn.q_proj	803,712	344,793,216
model.layers.13.self_attn.v_proj	114,816	344,908,032
model.layers.14.input_layernorm	896	344,908,928
model.layers.14.mlp.down_proj	4,358,144	349,267,072
model.layers.14.mlp.gate_proj	4,358,144	353,625,216
model.layers.14.mlp.up_proj	4,358,144	357,983,360
model.layers.14.post_attention_layernorm	896	357,984,256

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Module	Params	Running total
model.layers.14.self_attn.k_proj	114,816	358,099,072
model.layers.14.self_attn.o_proj	802,816	358,901,888
model.layers.14.self_attn.q_proj	803,712	359,705,600
model.layers.14.self_attn.v_proj	114,816	359,820,416
model.layers.15.input_layernorm	896	359,821,312
model.layers.15.mlp.down_proj	4,358,144	364,179,456
model.layers.15.mlp.gate_proj	4,358,144	368,537,600
model.layers.15.mlp.up_proj	4,358,144	372,895,744
model.layers.15.post_attention_layernorm	896	372,896,640
model.layers.15.self_attn.k_proj	114,816	373,011,456
model.layers.15.self_attn.o_proj	802,816	373,814,272
model.layers.15.self_attn.q_proj	803,712	374,617,984
model.layers.15.self_attn.v_proj	114,816	374,732,800
model.layers.16.input_layernorm	896	374,733,696
model.layers.16.mlp.down_proj	4,358,144	379,091,840
model.layers.16.mlp.gate_proj	4,358,144	383,449,984
model.layers.16.mlp.up_proj	4,358,144	387,808,128
model.layers.16.post_attention_layernorm	896	387,809,024
model.layers.16.self_attn.k_proj	114,816	387,923,840
model.layers.16.self_attn.o_proj	802,816	388,726,656
model.layers.16.self_attn.q_proj	803,712	389,530,368
model.layers.16.self_attn.v_proj	114,816	389,645,184
model.layers.17.input_layernorm	896	389,646,080
model.layers.17.mlp.down_proj	4,358,144	394,004,224
model.layers.17.mlp.gate_proj	4,358,144	398,362,368
model.layers.17.mlp.up_proj	4,358,144	402,720,512
model.layers.17.post_attention_layernorm	896	402,721,408
model.layers.17.self_attn.k_proj	114,816	402,836,224
model.layers.17.self_attn.o_proj	802,816	403,639,040
model.layers.17.self_attn.q_proj	803,712	404,442,752
model.layers.17.self_attn.v_proj	114,816	404,557,568
model.layers.18.input_layernorm	896	404,558,464
model.layers.18.mlp.down_proj	4,358,144	408,916,608
model.layers.18.mlp.gate_proj	4,358,144	413,274,752
model.layers.18.mlp.up_proj	4,358,144	417,632,896
model.layers.18.post_attention_layernorm	896	417,633,792
model.layers.18.self_attn.k_proj	114,816	417,748,608
model.layers.18.self_attn.o_proj	802,816	418,551,424
model.layers.18.self_attn.q_proj	803,712	419,355,136
model.layers.18.self_attn.v_proj	114,816	419,469,952
model.layers.19.input_layernorm	896	419,470,848
model.layers.19.mlp.down_proj	4,358,144	423,828,992
model.layers.19.mlp.gate_proj	4,358,144	428,187,136
model.layers.19.mlp.up_proj	4,358,144	432,545,280
model.layers.19.post_attention_layernorm	896	432,546,176
model.layers.19.self_attn.k_proj	114,816	432,660,992
model.layers.19.self_attn.o_proj	802,816	433,463,808
model.layers.19.self_attn.q_proj	803,712	434,267,520
model.layers.19.self_attn.v_proj	114,816	434,382,336
model.layers.20.input_layernorm	896	434,383,232
model.layers.20.mlp.down_proj	4,358,144	438,741,376
model.layers.20.mlp.gate_proj	4,358,144	443,099,520
model.layers.20.mlp.up_proj	4,358,144	447,457,664
model.layers.20.post_attention_layernorm	896	447,458,560
model.layers.20.self_attn.k_proj	114,816	447,573,376
model.layers.20.self_attn.o_proj	802,816	448,376,192
model.layers.20.self_attn.q_proj	803,712	449,179,904
model.layers.20.self_attn.v_proj	114,816	449,294,720

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Module	Params	Running total
model.layers.21.input_layernorm	896	449,295,616
model.layers.21.mlp.down_proj	4,358,144	453,653,760
model.layers.21.mlp.gate_proj	4,358,144	458,011,904
model.layers.21.mlp.up_proj	4,358,144	462,370,048
model.layers.21.post_attention_layernorm	896	462,370,944
model.layers.21.self_attn.k_proj	114,816	462,485,760
model.layers.21.self_attn.o_proj	802,816	463,288,576
model.layers.21.self_attn.q_proj	803,712	464,092,288
model.layers.21.self_attn.v_proj	114,816	464,207,104
model.layers.22.input_layernorm	896	464,208,000
model.layers.22.mlp.down_proj	4,358,144	468,566,144
model.layers.22.mlp.gate_proj	4,358,144	472,924,288
model.layers.22.mlp.up_proj	4,358,144	477,282,432
model.layers.22.post_attention_layernorm	896	477,283,328
model.layers.22.self_attn.k_proj	114,816	477,398,144
model.layers.22.self_attn.o_proj	802,816	478,200,960
model.layers.22.self_attn.q_proj	803,712	479,004,672
model.layers.22.self_attn.v_proj	114,816	479,119,488
model.layers.23.input_layernorm	896	479,120,384
model.layers.23.mlp.down_proj	4,358,144	483,478,528
model.layers.23.mlp.gate_proj	4,358,144	487,836,672
model.layers.23.mlp.up_proj	4,358,144	492,194,816
model.layers.23.post_attention_layernorm	896	492,195,712
model.layers.23.self_attn.k_proj	114,816	492,310,528
model.layers.23.self_attn.o_proj	802,816	493,113,344
model.layers.23.self_attn.q_proj	803,712	493,917,056
model.layers.23.self_attn.v_proj	114,816	494,031,872
model.norm	896	494,032,768